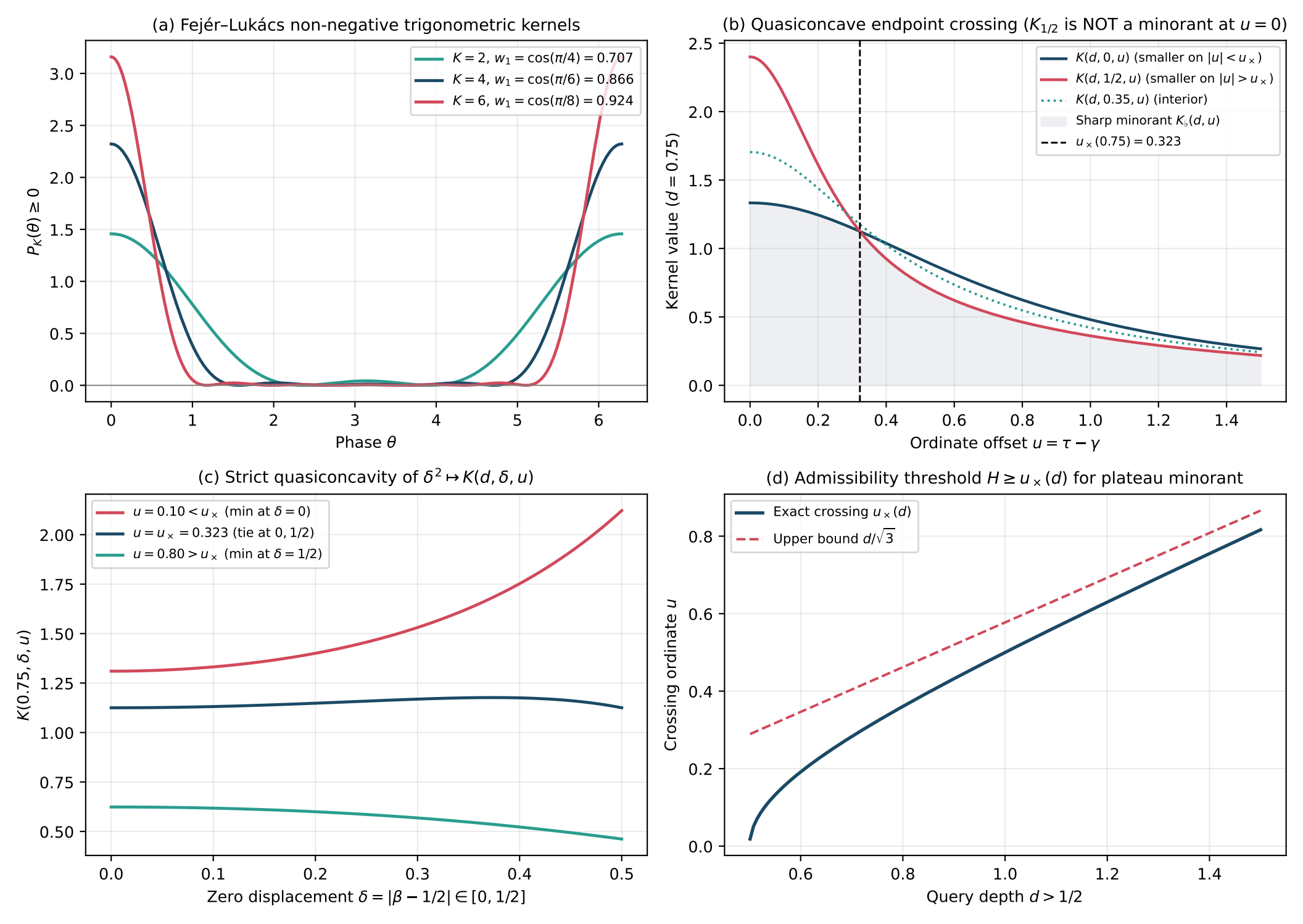
(a) Fejér–Lukács non-negative trigonometric kernels
$K = 2$, $w_1 = \cos(\pi/4) = 0.707$
$K = 4$, $w_1 = \cos(\pi/6) = 0.866$
$K = 6$, $w_1 = \cos(\pi/8) = 0.924$
$P_K(\theta) \geq 0$
Phase θ
(b) Quasiconcave endpoint crossing ($K_{1/2}$ is NOT a minorant at $u = 0$)
$K(d, 0, u)$ (smaller on $|u| < u_\times$)
$K(d, 1/2, u)$ (smaller on $|u| > u_\times$)
$K(d, 0.35, u)$ (interior)
Sharp minorant $K_\flat(d, u)$
$u_\times(0.75) = 0.323$
Kernel value ($d = 0.75$)
Ordinate offset $u = \tau - \gamma$
(c) Strict quasiconcavity of $\delta^2 \mapsto K(d, \delta, u)$
$u = 0.10 < u_\times$ (min at $\delta = 0$)
$u = u_\times = 0.323$ (tie at 0, 1/2)
$u = 0.80 > u_\times$ (min at $\delta = 1/2$)
$K(0.75, \delta, u)$
Zero displacement $\delta = |\beta - 1/2| \in [0, 1/2]$
(d) Admissibility threshold $H \geq u_\times(d)$ for plateau minorant
Exact crossing $u_\times(d)$
Upper bound $d/\sqrt{3}$
Crossing ordinate u
Query depth $d > 1/2$